

Cadi Ayyad University

Higher School of Technology of Safi

Computer Science Department • Web Techniques

Prof. Soufiane Hourri

Session 4 — CSS Layout, Boxes, and Better Page Design

Objective: Students master the CSS box model, selectors (class/id), and basic page layout.

1. Key Concepts

The CSS Box Model

Every HTML element is a rectangular box with four layers from inside to outside: **Content** (text/image), **Padding** (space inside the border), **Border** (the visible line), and **Margin** (space outside the border). The **total rendered width** of an element equals: width + left padding + right padding + left border + right border + left margin + right margin.

```
div {
  width: 300px;      /* content width */
  padding: 20px;     /* 20px on all sides */
  border: 2px solid #333;
  margin: 15px;
}
/* Total rendered width = 300 + 40 + 4 + 30 = 374px */
```

Classes and IDs

A **class** (.classname) can be applied to many elements. An **id** (#idname) must be unique on the page. Inline styles override internal/external styles; IDs override class rules; class rules override element rules — this is called CSS *specificity*.

```
/* Class: reusable */
.card { background-color: #fff; border: 1px solid #ccc; padding: 16px; }

/* ID: unique element */
#navbar { background-color: #1A3A5C; color: white; padding: 10px; }
```

Page Layout Zones

Concept	Meaning
header	Top zone: site title, navigation, logo.
main	Central zone: primary page content.
sidebar	Optional side zone: secondary info.

footer	Bottom zone: copyright, contact, links.
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2. Guided Examples

Example — Styled page layout with classes and IDs

```
/* CSS */
#header { background-color: #1A3A5C; color: white; text-align: center; padding: 20px; }
#footer { background-color: #333; color: #ccc; text-align: center; padding: 10px; }
.article { width: 600px; margin: 20px auto; padding: 20px; border: 1px solid #ddd; font-family: Arial; }
.highlight { background-color: yellow; font-weight: bold; }

<!-- HTML -->
<div id="header"><h1>My Blog</h1></div>
<div class="article">
  <h2>Post Title</h2>
  <p>Body text. <span class="highlight">Important word.</span></p>
</div>
<div id="footer"><p>Copyright 2026</p></div>
```

3. Exercises

Exercise 1 — Box Model Calculation

A developer writes the following CSS rule. Calculate the **total rendered width** of the element on screen, showing all steps of your calculation.

```
.panel {
  width: 400px;
  padding-left: 25px;
  padding-right: 25px;
  border: 3px solid black;
  margin-left: 40px;
  margin-right: 40px;
}
```

Calculation:

Total rendered width = _____ px

Now describe what changes if you add `box-sizing: border-box;` to this rule. How does the total rendered width change, and why?

Exercise 2 — Class vs ID — Specificity

Read the following CSS and HTML. Determine the **exact colour** that each element will display, and explain which rule wins and why.

```
/* CSS */
p          { color: black; }
.info      { color: blue; }
#warning   { color: red; }

<!-- HTML -->
<p>Normal paragraph.</p>
<p class="info">Info paragraph.</p>
<p id="warning" class="info">Warning paragraph.</p>
<p id="warning">Another warning.</p>
```

Paragraph	Colour displayed	Reason (which rule wins?)
'Normal paragraph'		
'Info paragraph'		
'Warning paragraph'		
'Another warning'		

Exercise 3 — Style a Page Section from a Description

Write complete CSS (internal <style> block) and matching HTML to produce a page that has exactly these visual characteristics:

- A header div with id page - header: dark navy background, white centred text, 30px padding.
- A main content area with class content - box: white background, 600px wide, centred (use margin: auto), 20px padding, 1px solid grey border.
- A footer div with id page - footer: dark grey background, light grey text, small font (12px), centred.
- Paragraphs inside the content box: Arial font, 15px size, 1.6 line-height.

Write your CSS and HTML below:

Exercise 4 — Read, Trace, and Describe

Study the CSS block below. For each numbered comment, write what visual change it produces.

```
.profile-card {  
  width: 250px;           /* 1 */  
  height: 300px;         /* 2 */  
  padding: 15px;         /* 3 */  
  border: 2px dashed #2E6DA4; /* 4 */  
  margin: 20px auto;     /* 5 */  
  background-color: #EAF2FB; /* 6 */  
  font-family: Arial, sans-serif; /* 7 */  
  border-radius: 8px;    /* 8 */  
}
```

#	What it does visually
1	
2	
3	
4	
5	
6	
7	
8	

Exercise 5 — Find and Correct — 4 CSS Errors

The stylesheet below has **4 errors** (syntax, value, and logic errors). Identify each one, state why it is wrong, and write the correct version.

```
#main-title {
  colour: #1A3A5C;
  text-align: centre;
  font-size: 24;
}

.sidebar {
  width: 30%;
  padding: 10px
  margin: auto;
  background: lightblue;
}
```

Error 1: _____

Error 2: _____

Error 3: _____

Error 4: _____

4. Session Recap

The CSS box model (content + padding + border + margin) determines the space every element occupies. Classes (.name) are reusable; IDs (#name) are unique. CSS specificity determines which rule wins when multiple rules target the same element (inline > ID > class > element). Using these tools, a page can be organised into clear zones: header, main, sidebar, footer. CSS makes pages clearer, more attractive, and easier to maintain.